

International Islamic University Chittagong (IIUC)

Department of Computer and Communication Engineering

Midterm Examination

Program: B.Sc (Engg.)
Course Code: CCE-3507
Total Marks: 30

Semester: Spring 2026
Course Title: System Analysis, Design and Development
Time: 1 Hour 30 Minutes

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
(ii) Course Outcomes (COs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

- CLO1** Analyze the nature of Information in systems analysis and design and its various components.
CLO2 Determine feasibility and handle managerial issues like data input and output related to the design.

Bloom's Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create
Q1. a) Define System. Explain how System Analysis and System Design are correlated.					CLO1	U 4
b) What is an information system? Discuss different types of information systems.					CLO1	R 6
Q2. a) A company is planning to develop a new information system but is facing difficulties in clearly identifying user requirements. The project team decides to arrange a series of structured meetings involving users, managers, and system analysts.					CLO2	U 5
Identify the approach being used and explain how it helps in effective requirement gathering with its steps.						
b) What is the Agile Model in software development? Explain its main features and advantages.					CLO1	An 5
Q3. a) Classify different types of costs within a software company.					CLO2	R 3
b) A reputed tech company is considering a project that involves large initial investment in Year 0, followed by benefits over the next five years.					CLO2	Ap/E 7

Year	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Total Benefits		45,000	50,000	47,000	41,000	75,000
Total Costs	100,000	10,000	12,000	16,000	12,700	18,000

- i. Calculate the ROI.
- ii. Perform cash flow analysis to Determine Break-Even Point.

Or

Q3. a) What is prototype in System Analysis and Design? Explain its purpose in the system development process.

CLO2

R

3

b) "Y" group of companies wants to introduce a new computer system for its new Warehouse. The costs and benefits are as follows:

CLO2

Ap/E

7

Year	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Total Benefits		45,000	50,000	55,000	61000	59000
Total Costs	100,000	10,000	13,000	16,000	13000	17500

- i. Given a discount rate of 8%, perform present value analysis on the data.
- ii. What is your recommendation for the company?

International Islamic University Chittagong (IIUC)

Department of Computer and Communication Engineering

Midterm Examination

Program: **B.Sc (Engg.)**
 Course Code: **CCE-3507**
 Total Marks: **30**

Semester: **Autumn 2025**
 Course Title: **System Analysis, Design and Development**
 Time: **1 Hour 30 Minutes**

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
 (ii) Course Outcomes (COs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

- CLO1** Analyze the nature of information in systems analysis and design and its various components.
CLO2 Determine feasibility and handle managerial issues like data input and output related to the design.

Bloom's Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create
Q1. a) Define information. Write the essential qualities of information.					CLO1	U 2
b) What is an information system? Discuss different types of information systems.					CLO1	R 5
c) What qualities should a person have for the role of a system analyst in an information system?					CLO1	An 3
Q2. a) Differentiate between waterfall model and agile model.					CLO1	U 4
b) Explain the System Development Life Cycle (SDLC) with its stages, listing activities performed and outputs produced in each stage.					CLO1	R/An 6
Q3. a) Define feasibility study in system analysis and design. Discuss its importance before beginning system development.					CLO2	An 3
b) 'X' company is considering a project that involves large initial costs in Year 0, followed by benefits over the next three years.					CLO2	Ap/E 7

Year	Year 0	Year 1	Year 2	Year 3
Total Benefits		45,000	50,000	57,000
Total Costs	100,000	10,000	12,000	16,000

- i. Calculate the ROI.
- ii. Perform cash flow analysis to Determine Break-Even Point.(use payback method)

Or

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Department of Computer and Communication Engineering

Midterm Examination

Program: **B.Sc. (Engg.)**
 Course Code: **CCE-3507**
 Total Marks: **30**

Semester: **Spring-2025**
 Course Title: **System Analysis, Design and Development**
 Time: **1 Hours 30 Minutes**

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
 (ii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions									
CLO1	Analyse the nature of information in systems analysis and design and its various components.								
CLO2	Determine feasibility and handle managerial issues like data input and output related to the design.								
Bloom's Levels of the Questions									
Letter Symbols	R	U	Ap	An	E	C			
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create			
Q.1	a)	Define the different types of information and provide relevant examples for each. Why do we need a computer based information system?				CLO1	U	2	
	b)	What are the essential qualities of information? List and define three key roles a systems analyst plays throughout the system development process.				CLO1	U	2	
	c)	Illustrate the systems development life cycle (SDLC) with its seven phases, listing activities performed and outputs produced in each phase.				CLO1	R/U	6	
Q.2	a)	As part of your systems analysis project to update the <i>automated accounting functions</i> for a reputed university, you will interview Mr. Masud Ahmed, the Director of Accounts. Write four to six interview objectives covering his use of information sources, information formats, decision-making frequency, desired qualities of information, and decision-making style.				CLO1	Ap	5	
	(i) In a paragraph, write down how you will approach Mr. Masud to set up an interview.								
	(ii) State which structure you will choose for this interview. Why?								
	(iii) Mr. Masud has four subordinates who also use the system. Would you interview them also? Why or why not?								
	(iv) Would you also try to interview students? Are there better ways to get the opinions of students? Why or why not?								
	b)	What kinds of information is a systems analyst seeking through the use of questionnaires or surveys? List four situations in which the use of questionnaires is appropriate. What are the seven guidelines for choosing language for a questionnaire?				CLO1	R/U	5	
Q.3	a)	What are the three ways to find out about problems or opportunities that might call for a systems solution?				CLO2	U/An	3	
	b)	Interglobal Paper Company has asked for your help in comparing its present computer system with a new one its board of directors would like to see implemented. The proposed system costs and present system costs are as given in Fig. 3(b).				CLO2	Ap/E	7	

Item	Proposed System Costs	Present System Costs
Year 1		
Equipment lease	\$20,000	\$11,500
Salaries	30,000	50,000
Overhead	4,000	3,000
Development	30,000	—
Year 2		
Equipment lease	\$20,000	\$10,500
Salaries	33,000	55,000
Overhead	4,400	3,300
Development	12,000	—
Year 3		
Equipment lease	\$20,000	\$10,500
Salaries	36,000	60,000
Overhead	4,900	3,600
Development	—	—
Year 4		
Equipment lease	\$20,000	\$10,500
Salaries	39,000	66,000
Overhead	5,500	4,000
Development	—	—

Fig. 3(b)

The following are system benefits for Interglobal Paper Company:

Year	Benefits
1	\$55,000
2	75,000
3	80,000
4	85,000

(i) Use the costs of Interglobal Paper's proposed system to determine the payback period. (Use the *payback method*.)

(ii) Graph the benefits versus the costs and indicate the payback period.

OR

Q.3 a) Define tangible costs and tangible benefits. List the steps involved in conducting a feasibility analysis for a project. **CLO2 U/An 3**

b) Hoque Foods of Chattogram wants to introduce a new computer system for its perishable products warehouse. The costs and benefits are as follows: **CLO2 Ap/E 7**

Year	Costs	Benefits
1	\$33,000	\$21,000
2	34,600	26,200
3	36,300	32,700
4	38,100	40,800
5	40,000	51,000
6	42,000	63,700

(i) Given a discount rate of 13 percent (0.13), perform *present value analysis* on the data for Hoque Foods.

(ii) What is your recommendation for Hoque Foods?

International Islamic University Chittagong
Department of Computer and Communication Engineering

B. Sc. Engineering in CCE

Class Test, Autumn 2024

Course Code: CCE-3507

Course Title: System Analysis, Design and Development

Time: 45 Min.

Full Marks: 15

(i) Answer all of the questions. The figures in the right-hand margin indicate full marks.

(ii) Course Outcomes (COs) and Bloom's Levels are mentioned in additional Columns.

Course Outcomes (COs) of the Questions

CLO1 Analyse the nature of Information in systems analysis and design and its various components.

CLO2 Determine feasibility and handle managerial issues like data input and output related to the design.

Bloom's Levels of the Questions

Letter Symbols Meaning			R Remember	U Understand	Ap Apply	An Analyze	E Evaluate	C Create	
Q.1	a)	List the benefits of applying systems analysis and design techniques when approaching computerized information systems.					CLO1	U	1
	b)	What personal attributes are helpful to a systems analyst? Provide a definition for each one.					CLO1	U	4
Q.2	a)	Illustrate the systems development life cycle (SDLC) along with seven phases. List the activities performed and the outputs produced during the different phases of this cycle.					CLO1	R/ U	5
Q.3	a)	List the five steps in interview preparation. Define what is meant by open-ended interview questions. List three benefits and drawbacks of using them.					CLO1	An	5
OR									
Q.3	a)	Define what is meant by funnel structure of questions arrangement in interviews. When is it useful to employ this structure in interviews?					CLO1	An	5

International Islamic University Chittagong
Department of Computer and Communication Engineering

B. Sc. Engineering in CCE

Midterm Examination, Spring 2024

Course Code: CCE-3507

Course Title: System Analysis, Design and Development

Time: 1 Hour 30 Minute

Full Marks: 30

(i) Answer all of the questions. The figures in the right-hand margin indicate full marks.

(ii) Course Outcomes (COs) and Bloom's Levels are mentioned in additional Columns.

Course Outcomes (COs) of the Questions

CLO1 Analyse the nature of information in systems analysis and design and its various components.

CLO2 Determine feasibility and handle managerial issues like data input and output related to the design.

Bloom's Levels of the Questions

	Letter Symbols	Meaning	Bloom's Levels of the Questions					
			R	U	Ap	An	E	C
			Remember	Understand	Apply	Analyze	Evaluate	Create
Q.1	a)	List the advantages of using systems analysis and design techniques in approaching computerized information systems.						CLO1 U 2
	b)	What are the essential qualities of information? A large hospital wants to streamline patient care and improve medical record access. What types of operational, tactical and strategic information are needed to develop an information system that facilitates seamless communication and record access?						CLO1 An 4
	c)	List the situations that warrant use of joint application design (JAD) in place of personal organizational interviews. List the potential benefits and drawbacks of using JAD as an alternative to personal interviews.						CLO1 R 4
Q.2	a)	Illustrate the systems development life cycle (SDLC) along with seven phases. List the activities performed and the outputs produced during the different phases of this cycle.						CLO1 R/U 6
	b)	You're gathering information about customer satisfaction with a service. Design a questionnaire using both open-ended and closed questions so that they follow good practice and result in useful information for the systems analysts.						CLO1 Ap/An 4
Q.3	a)	How would you determine whether the development of a proposed project is feasible or not? Discuss in brief. Write down the steps followed in feasibility analysis of a project.						CLO2 An 5
	b)	GEC Electronics, a small company, has set up a computer service. The table that follows shows the revenue expected for the first five months of operation, in addition to the costs for office remodeling and so on. Determine the cash flow and accumulated cash flow for the company. When is GEC's expected to show a profit?						CLO2 An 5

Table 3(b)

	July	August	September	October	November
Revenue	\$35,000	\$36,000	\$42,000	\$48,000	\$57,000
Costs					
Office remodeling	\$25,000	\$8,000			
Salaries	11,000	12,100	\$13,300	\$14,600	\$16,000
Training	6,000	6,000			
Equipment lease	8,000	8,480	9,000	9,540	10,110
Supplies	3,000	3,150	3,300	3,460	3,630

OR

- Q.3 a)** What are the three ways to find out about problems or opportunities that might call for a systems solution? Discuss in brief. **CLO2 An 3**
- b)** T. K. Paper Company has asked for your help in comparing its present computer system with a new one its board of directors would like to see implemented. The proposed system costs and present system costs are as given in Fig. 3(b). **CLO2 An 7**

Item	Proposed System Costs	Present System Costs
Year 1		
Equipment lease	\$20,000	\$11,500
Salaries	30,000	50,000
Overhead	4,000	3,000
Development	30,000	—
Year 2		
Equipment lease	\$20,000	\$10,500
Salaries	33,000	55,000
Overhead	4,400	3,300
Development	12,000	—
Year 3		
Equipment lease	\$20,000	\$10,500
Salaries	36,000	60,000
Overhead	4,900	3,600
Development	—	—
Year 4		
Equipment lease	\$20,000	\$10,500
Salaries	39,000	66,000
Overhead	5,500	4,000
Development	—	—

Fig. 3(b)

The following are system benefits for T. K. Paper Company:

Year	Benefits
1	\$55,000
2	75,000
3	80,000
4	85,000

- (i) Using the break-even analysis technique, determine the year in which T. K. Paper will break-even.
- (ii) Use the costs of T. K. Paper's proposed system to determine the payback period. (Use the payback method.)
- (iii) Graph the benefits versus the costs and indicate the payback period.

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Course Code: **CCE-3507**

Total Marks: **30**

Semester: **Spring 2023**

Course Title: **System Analysis, Design and Development**

Time: **1 Hour 30 Minutes**

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
 (ii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

CLO1 Analyse the nature of information in systems analysis and design and its various components.

CLO2 Determine feasibility and handle managerial issues like data input and output related to the design.

1. a) Define various types of information with relevant examples. Why do we need a computer-based information system? **CLO1** **U** **3**
 b) What personal qualities are helpful to a systems analyst? List them. **CLO1** **U** **2**
 c) When are open-ended questions appropriate for use in interviewing? List eight benefits and five drawbacks of using open-ended questions. **CLO2** **R/U** **5**
 Or,
 List the situations that warrant use of joint application design (JAD) in place of personal organizational interviews. List the potential benefits and drawbacks of using JAD as an alternative to personal interviews.
2. a) What are the essential qualities of information? List various types of information systems for users that the systems analysts recommend, design, and maintain. List the different tools that are used by the systems analyst. **CLO1** **R/U** **4**
 b) List all of the activities performed and the outputs produced in the seven phases of the systems development life cycle (SDLC). **CLO1** **R/U** **6**
3. a) As a member of systems analysis team, you will interview the local manager of **LOWCO**, an outlet of a national discount chain, who has asked you to work on a management information system to provide inventory information. Write a series of six *closed questions* that cover the subject of decision-making style for the manager. **CLO2** **Ap/E** **4**

b) **CLO2** **E** **6**

Year	Costs	Benefits
1	\$ 33,000	\$ 21,000
2	34,600	26,200
3	36,300	32,700
4	38,100	40,800
5	40,000	51,000
6	42,000	63,700

Alamo Foods of San Antonio wants to introduce a new computer system for its perishable products warehouse. The costs and benefits are as stated in the above table:

- (i) Given a discount rate of 13 percent (0.13), perform present value analysis on the data for Alamo Foods.
 (ii) What is your recommendation for Alamo Foods?

OR

3. a) As a member of systems analysis team, you will interview the local manager of LOWCO, an outlet of a national discount chain, who has asked you to work on a management information system to provide inventory information. Write a series of six *open-ended questions* that cover the subject of decision-making style for the manager.
- b) Interglobal Paper Company has asked for your help in comparing its present computer system with a new one its board of directors would like to see implemented. The proposed system costs and present system costs are as given in Fig. 3(b).

CLO2 Ap 4
/E

CLO2 Ap 6
/E

Item	Proposed System Costs	Present System Costs
Year 1		
Equipment lease	\$20,000	\$11,500
Salaries	30,000	50,000
Overhead	4,000	3,000
Development	30,000	—
Year 2		
Equipment lease	\$20,000	\$10,500
Salaries	33,000	55,000
Overhead	4,400	3,300
Development	12,000	—
Year 3		
Equipment lease	\$20,000	\$10,500
Salaries	36,000	60,000
Overhead	4,900	3,600
Development	—	—
Year 4		
Equipment lease	\$20,000	\$10,500
Salaries	39,000	66,000
Overhead	5,500	4,000
Development	—	—

Fig. 3(b)

The following are system benefits for Interglobal Paper Company:

Year	Benefits
1	\$55,000
2	75,000
3	80,000
4	85,000

(i) Use the costs of Interglobal Paper's proposed system to determine the payback period. (Use the payback method.)

(ii) Graph the benefits versus the costs and indicate the payback period.

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Semester: **Spring 2022**
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 Time: **1 Hour 30 Minutes**

(I) Answer all of the questions. The figures in the right-hand margin indicate full marks.						
(II) Course Outcomes (COs) and Bloom's Levels are mentioned in additional columns.						
Course Outcomes (COs) of the Questions						
CLO1	Analyse the nature of Information in systems analysis and design and its various components.					
CLO2	Determine feasibility and handle managerial issues like data input and output related to the design.					
Bloom's Levels of the Questions						
Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

- Q.1 a) List three roles that a systems analyst is called upon to play. Provide a definition for each one. CLO1 U 3
- b) What are the three ways to find out about problems or opportunities that might call for a systems solution? CLO2 An/ 2
E
- c) As part of your systems analysis project to update the *automated accounting functions* for a reputed university, you will interview Mr. Masud Ahmed, the Director of Accounts. Write four to six interview objectives covering his use of information sources, information formats, decision-making frequency, desired qualities of information, and decision-making style. CLO1 Ap/ 5
E
- (i) In a paragraph, write down how you will approach Mr. Masud to set up an interview.
- (ii) State which structure you will choose for this interview. Why?
- (iii) Mr. Masud has four subordinates who also use the system. Would you interview them also? Why or why not?
- (iv) Would you also try to interview students? Are there better ways to get the opinions of students? Why or why not?
- Q.2 a) Briefly describe the seven phases of the systems development life cycle (SDLC) along with the diagram. CLO1 R/ 6
U
- b) What is a *probing* question? What is the purpose of using a *probing* question in interviews? What are the seven guidelines for choosing language for a *questionnaire*? CLO1 R/ 4
U
- Q.3 a) Figure 3(a) is a questionnaire designed by an employee of Green Toe Textiles, which specializes in manufacturing men's socks. Di Wooly wrote the questionnaire because, as the office manager at headquarters in Juniper, Tennessee, she is concerned with the proposed purchase and implementation of a new computer system. CLO1 Ap/ 5
E

Hi! All Employees

- What's new? According to the grapevine, I hear we're in for a new computer. Here are some questions for you to think about.
- How long have you used the old computer? _____
 - How often does it go down? _____
 - Who repairs it for you? _____
 - When was the last time you suggested a new improvement to the computer system and it was put into use? What was it? _____
 - When was the last time you suggested a new improvement to the computer system and nobody used it? What was it? _____
 - Do you use a VDT or printer or both? _____
 - How fast do you type? _____
 - How many people need to access the database regularly at your branch? Is there anyone not using the computer now who would like to? _____

Fig. 3(a): A questionnaire developed by Di Wooly.

Based on what you assume Ms. Wooly is trying to get through the questionnaire, rewrite and reorder the questions (use both open-ended and closed questions) so that they follow good practice and result in useful information for the systems analysts. Indicate next to each question that you write whether it is open-ended or closed.

- b) T. K. Paper Company has asked for your help in comparing its present computer system with a new one its board of directors would like to see implemented. The proposed system costs and present system costs are as given in Fig. 3(b).

Item	Proposed System Costs	Present System Costs
Year 1		
Equipment lease	\$20,000	\$11,500
Salaries	30,000	50,000
Overhead	4,000	3,000
Development	30,000	—
Year 2		
Equipment lease	\$20,000	\$10,500
Salaries	33,000	55,000
Overhead	4,400	3,300
Development	12,000	—
Year 3		
Equipment lease	\$20,000	\$10,500
Salaries	36,000	60,000
Overhead	4,900	3,600
Development	—	—
Year 4		
Equipment lease	\$20,000	\$10,500
Salaries	39,000	66,000
Overhead	5,500	4,000
Development	—	—

Fig. 3(b)

- (i) Using the *break-even analysis* technique, determine the year in which T. K. Paper will break even.
(ii) Graph the costs and show the break-even point.

OR

- Q.3 a) *GEC Electronics*, a small company, has set up a computer service. The table that follows shows the revenue expected for the first five months of operation, in addition to the costs for office remodeling and so on. Determine the cash flow and accumulated cash flow for the company. When is *GEC's* expected to show a profit? CLO2 Ap/ 5
E

	July	August	September	October	November
Revenue	\$35,000	\$36,000	\$42,000	\$48,000	\$57,000
Costs					
Office remodeling	\$25,000	\$8,000			
Salaries	11,000	12,100	\$13,300	\$14,600	\$16,000
Training	6,000	6,000			
Equipment lease	8,000	8,480	9,000	9,540	10,110
Supplies	3,000	3,150	3,300	3,460	3,630

- b) Hoque Foods of Chattogram wants to introduce a new computer system for its perishable products warehouse. The costs and benefits are as follows: CLO2 Ap/ 5
E

Year	Costs	Benefits
1	\$33,000	\$21,000
2	34,600	26,200
3	36,300	32,700
4	38,100	40,800
5	40,000	51,000
6	42,000	63,700

- (i) Given a discount rate of 12 percent (0.12), perform *present value analysis* on the data for Hoque Foods.
(ii) What is your recommendation for Hoque Foods?